جامعة خليفة
Khalifa University

Department of Computer Science (COSC)

Introduction to Software Engineering (COSC336)

Spring 2026

BookMyManzil

Date: 26/01/2026

Prepared by:
Tehsin Shaik 100064806

Prepared For: Khalifa University

Introduction

I. Overview of the project concept

The hospitality industry is evolving rapidly, and digital systems are becoming essential for managing everything from reservations to guest services, pricing, and daily hotel operations. While many hotels have management systems to handle basic tasks like room bookings and billing, these systems often lack intelligence, personalization, and real-time adaptability. As guest expectations continue to rise, there is a growing need for smarter systems that can enhance operational efficiency while delivering a more personalized guest experience.

This project focuses on the development of a Smart, AI-Driven Hotel Booking and Guest Experience Management System that integrates traditional hotel management functionalities with AI-powered enhancements. The system is designed to support both guests and hotel staff by automating core operations, improving decision-making through data analysis, and enabling natural language interaction through an AI-based virtual assistant. By combining classic software engineering principles with emerging AI technologies, the project addresses current limitations in hotel management systems and reflects real-world industry needs.

II. Purpose and objectives of the document

The purpose of this document is to present the "Initial Plan and Requirement Gathering" for Phase 1 of the project. It provides a structured overview of the proposed system, including the project scope, objectives, key stakeholders, and initial functional and non-functional requirements. Altogether, this document establishes the foundation for all subsequent phases of the software development lifecycle by defining the system's direction, constraints, and development approach.

Project Overview

I. High-level description of the proposed system or software

The proposed system is a Smart, AI-Driven Hotel Booking and Guest Experience Management System designed to support hotel operations while enhancing guest interaction through intelligent automation. It provides core hotel management functionalities, including room booking and availability management, digital check-in and check-out processing, guest service request handling, and secure user account management with role-based access control. These features support daily hotel operations and improve coordination between guests and staff.

In addition to traditional functionalities, the system integrates AI-enhanced features such as a conversational virtual assistant for natural language interaction and analytics modules for demand forecasting and dynamic pricing. These capabilities enable personalized guest experiences, data-driven decision-making, and improved operational efficiency.

II. Background and Context of the Project

Many existing hotel management systems primarily support basic functions such as room reservations and billing, often relying on manual processes for guest services and static pricing rules. This limits their ability to respond quickly to changing demand and guest expectations, leading to operational inefficiencies, slower service and reduced personalization.

Advances in data analysis and AI technologies provide opportunities to address these limitations. By using historical booking data, adaptive pricing strategies, and natural language interaction, modern systems can improve efficiency and service quality. This project is developed within this context, demonstrating how AI-enhanced features can make hotel management systems more flexible, responsive, and guest-centric.

III. Business Goals and Objectives

The main goal of this project is to improve hotel operational efficiency while enhancing guest satisfaction. By automating routine processes and centralizing information, the system aims to reduce manual effort, minimize errors, and improve response times to guest requests.

From an academic perspective, the project aims to apply core software engineering concepts in a practical setting. This includes designing a modular system, following best practices throughout the development lifecycle, and exploring how AI technologies can be responsibly integrated into real-world software systems.

IV. Project Constraints (Budget, Timeline, Resources)

The project is developed within a fixed academic timeline of approximately 11 weeks, with clearly defined phases and submission deadlines. These time constraints influence the project scope, requiring careful prioritization of features to ensure timely delivery.

As an academic project, no financial budget is allocated. Development relies on open-source tools, freely available frameworks, and accessible datasets. Additionally, limited team resources require a focus on essential functionalities and avoidance of unnecessary complexity. Feature selection and design decisions are made with feasibility and practicality in mind.

To effectively manage these constraints, an Agile development approach is used. This supports iterative development, allowing features to be refined gradually and adjustments to be made when needed. It also encourages continuous review and improvement throughout the project timeframe, helping ensure that the final prototype remains realistic, well-structured, and aligned with both academic requirements and real-world use cases.

V. Scope of the Project

The scope of this project focuses on designing and prototyping a functional hotel management system that includes both core features and selected AI-driven enhancements. This includes room booking, guest check-in and check-out, service request management, user role handling, AI-based guest interaction, and basic analytics for pricing and demand forecasting.

The project does not aim to deliver a fully commercial system. Large-scale deployment, advanced integrations, and extensive real-world data usage are considered out of scope. Instead, the emphasis is on building a realistic and well-structured prototype that reflects real hotel operations and meets the objectives of the course.

Stakeholder Analysis

Table 1 provides an overview of the identified stakeholders and highlights their roles, responsibilities, and expectations.

Stakeholders	Roles and Responsibilities	Needs and Expectations
Guests	• Search for different options • Make reservations • Check-in/check out • Request additional services (maintenance, housekeeping, room service)	• Variety of opportunities • User-friendly system design • Efficient digital check-in/check-out • Personalized recommendations and assistance (through AI assistant) • Personal data and payment security
Travel agencies (and tour companies)	• Search for best options for their clients • Check availability • Make reservations	• Access to room availability and pricing information • Support for group reservations • Easy communication to manage changes and cancellations
Technical staff	• Ensure security • Handle system updates • Fix errors • System maintenance	• Ease of maintenance and opportunities for future enhancements • Secure authentication • Stable system architecture
Front desk (reception) staff	• Manage and verify guest bookings • Perform check-in/check-out • Handle guest inquires	• Efficient reservation verification • Real-time booking and availability information • Visibility of guest status (checked-in, checked-out) • Easy to learn how to use system • Efficient (especially in peak hours) system

		• Less human/operational errors and paperwork
Service staff (housekeeping, maintenance, room service)	• Receive requests from guests • Manage requests • Update the status of requests	• Prioritized and well-organized task lists • Timely notifications of requests • Ability to update the status of requests • Easy to use system
Hotel managers	• Monitor system and hotel performance • Make decisions regarding hotel operations based on system insights • Analyze reports and data	• Reliable analytics of occupancy, revenue and overall operation of the hotel • AI-driven pricing recommendations and demand forecasting

Table 1: Stakeholder Analysis

Requirements Gathering

To gather the requirements, a combination of qualitative and analytical techniques was used. Surveys and questionnaires were conducted to collect user feedback regarding booking experience, service quality, and desired personalization features. In addition, interviews with key stakeholders such as hotel administrators, reception staff, and service staff were considered to understand operational workflows, challenges, and expectations from the system. Existing online hotel booking and management platforms were also reviewed to identify commonly offered features and recurring issues highlighted in user reviews, helping to understand industry best practices and user expectations.

The gathered requirements were categorized into functional and non-functional requirements, where functional requirements include both core hotel management functionalities and AI-enhanced system functionalities. The requirements were then prioritized based on operational necessity and user impact, with essential hotel operations classified as high priority, AI-enhanced functionalities classified as medium priority, and optional future enhancements classified as lower priority.

Initial Requirements Specification

1. Functional Requirements

1.1 Room Booking and Availability Management

- The system will allow users to search for available rooms across multiple hotels based on selected dates, room type, and occupancy.
- The system will display room availability and pricing information in real time.

- The system will prevent booking conflicts by ensuring that a room cannot be reserved for overlapping dates.
- The system will allow guests to create, view, modify, and cancel reservations.
- The system will allow guests to complete reservations using online payment methods.

1.2 User Account and Role Management

- The system will support user accounts for different roles, including guests, reception staff, service staff, and hotel administrators.
- The system will authenticate users before granting access to system functionalities.
- The system will enforce role-based access control, restricting features according to user roles.

1.3 Guest Check-In and Check-Out Processing

- The system will allow reception staff to verify guest reservations during the check-in process.
- The system will update room status (e.g., available, occupied) during check-in and check-out.
- The system will maintain the current status of guests (e.g., checked-in, checked-out).

1.4 Guest Service Request Handling

- The system will allow guests to submit service requests, including housekeeping, maintenance, and room service.
- The system will allow service staff to view assigned service requests.
- The system will allow service staff to update the status of requests (e.g., pending, in progress, completed).
- The system will notify relevant staff when new service requests are submitted.
- The system will notify guests about updates to the status of their service requests.

1.5 Administrative and Management Functions

- The system will allow hotel administrators to manage room information, pricing, and availability.
- The system will allow administrators to manage staff accounts and assign roles.
- The system will provide access to basic reports related to bookings and hotel operations.

1.6 AI-Enhanced Functions

- The system will provide a conversational virtual assistant to support guests with booking inquiries and service requests.

- The system will allow guests to interact with the virtual assistant using natural language.
- The system will analyze historical booking data to support demand forecasting.
- The system will provide pricing recommendations based on demand trends and historical data.

2. Non-Functional Requirements

- The system should ensure security and compliance by protecting user data, enforcing role-based access control, supporting secure online payment methods, and complying with applicable UAE government regulations related to data protection and online services.
- The system should provide a user-friendly interface suitable for both guests and hotel staff, requiring minimal training.
- The system should deliver reliable and timely performance, providing fast responses to user actions and operating consistently without data loss.

Initial Project Plan Outline

To represent the project timeline, a Gantt chart was created to present the complete project timeline, showing the sequence and duration of each phase. The project will take us approximately 11 weeks to complete. The most important deadlines are the draft implementation with the demo (April 20th) and the final submission and presentation (April 27th).

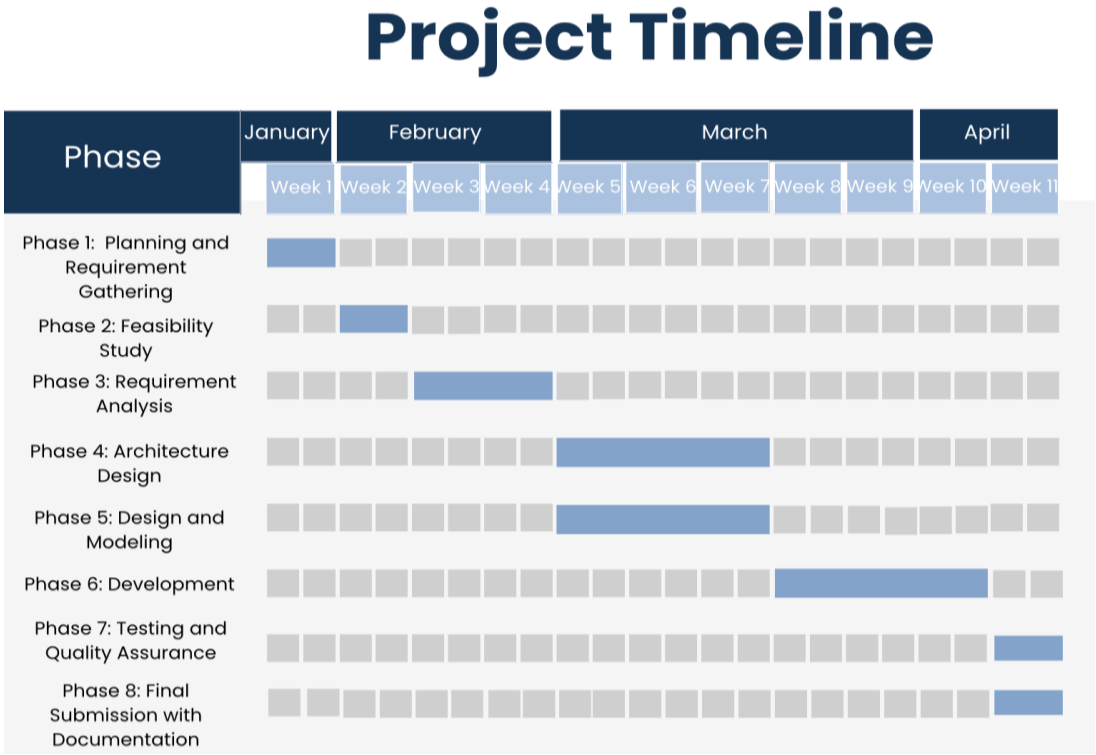

Chart 1: Gantt Chart showing Project Timeline

The team decided to use the Agile development method to support iterative progress and continuous feedback after each phase, allowing issues to be identified and addressed early.

Table 2 shows the resource allocation plan, outlining the team members, their assigned roles, and key responsibilities. Due to the small team size, each member takes on multiple roles to distribute the workload evenly and ensure all project activities are adequately covered throughout the development lifecycle.

Team Member	Roles	Responsibilities
Tehsin Shaik	Project Manager (PM), Backend Developer & AI Integration Lead	Coordinates project activities and phase deliverables; leads backend development and database design; oversees and contributes to AI integration features.
Mariam AlQubaisi	Requirements Analyst, QA Tester, Backend Developer & AI Support	Gathers and documents functional and non-functional requirements; contributes to backend development and business logic implementation; designs and executes test cases; supports AI integration by validating outputs and testing system behavior.
Ekaterina Gorkova	Frontend Developer, UI Designer, Backend Developer & AI Support	Designs and implements the user interface; develops frontend components; supports AI integration; and contributes to backend development by ensuring consistency between frontend and backend functionality.

Table 2: Resource Allocation Plan